

Democracy Bench: Can a Language Model Be Steered Toward the Public Without Eroding Democratic Rights?

Anonymous submission

Abstract

Civil servants and public bodies are meant to track the opinions and preferences of their constituents, and democracies have built mechanisms for exactly that: elections, consultations, and opinion polls all filter the public’s views into policy. It is far less clear whose opinions and values the AI models now entering government reflect. We introduce **Democracy Bench**, a twofold evaluation harness. It first measures how closely a model tracks public opinion, built on two waves of the British Social Attitudes survey, while separating fluid democratic *preferences* (tax, public spending, healthcare) from the democratic *rights* that should hold regardless (due process, the right to appeal, free expression). It then attempts to steer models toward, and away from, the public’s preferences through a five-rung ladder of methods, from prompting to fine-tuning. Every rung fails but one. Evidence in context lets a model track the real 2022→2024 shift in public attitudes, yet the same deference lets whoever writes that context push a model over its rights floors, and a direct instruction destroys them almost entirely. The pattern holds across three tiers of model, from the local quantized checkpoints we build the harness on, through API-hosted models of six families, to the frontier open-weight checkpoints that sovereign-AI programmes propose to fine-tune; one of those ships below a rights floor before any attack reaches it. The guard that remains is architectural. Floor items never see public-opinion evidence at all, so the one demonstrated attack cannot reach them. Every reported number regenerates from released artifacts. Democracy Bench reframes the question from whether a model can be steered to who controls the channel that steers it, and lays groundwork for legitimate public participation in the models deployed in the public’s name.

1 Introduction

Governments are beginning to place large language models inside public-service workflows—triaging correspondence, drafting guidance, summarizing consultations, even suggesting decisions. Civil servants and the public bodies they serve are meant to track the opinions and preferences of their constituents, and democracies have built mechanisms for exactly that; elections, consultations, and opinion polls all let the

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

public’s opinion filter into government. There is no comparable channel into a model’s parameters. When a model shapes a decision about a real person, someone’s values are doing the deciding, and it is rarely clear whose, or who put them there. The recurring hope is that a model can act as a *policy delegate*, an agent that reflects what a real public wants. Whether a deployed model can be *governed* that way, and by whom, is an empirical question existing evaluations do not answer.

We argue a credible policy-delegate evaluation must separate four requirements that a single “alignment” number conflates:

1. **Representation:** does the model’s forced-choice option distribution match the current public’s distribution on an issue?
2. **Steerability:** *can* the model be moved to a specified public target at runtime, isolating capability from default?
3. **Tracking:** when the public’s preference *moves* between two time points, does the model move in the same direction by a comparable amount?
4. **Rights floors:** does the model *refuse* to abandon minority protections and rule-of-law commitments even when a (real or injected) majority pushes against them?

Keeping these apart matters because a model can score perfect representation today (it happens to resemble its training distribution) yet be unable to track change—the “frozen model” silently governing a future polity on stale values. And the mirror risk is *majoritarian capture*: a delegate so eager to follow the public that it will erode a right the moment a majority—or an adversary posing as one—asks it to. Democratic theory has long separated the preferences a majority may set from the protections it may not vote away (Dahl 1989), and that line, between “follow the electorate” and “obey the majority against a minority,” is the single most important design decision in the benchmark. We build it in from the start as a *legitimacy boundary* between two item classes, and we run the experiment in three tiers: we build the evaluation cheaply on local quantized models, prove the harness generalizes across API-hosted model families, and then aim it at the frontier open-weight checkpoints that sovereign-AI programmes propose to fine-tune.

Contributions. (1) **Democracy Bench**, a judge-free, survey-grounded harness that scores representation, steer-

ability, tracking, and rights-floor stability against British Social Attitudes (BSA) microdata, with a *tracking elasticity* metric (§3). (2) A systematic evaluation of a five-rung *steering ladder*, the benchmark’s interventional arm: in-context conditioning dominates, decode/activation/weight levers are null or destructive, and we characterize *why* each rung fails (§4–§5). (3) A decomposed attack surface, established by a preregistered channel experiment: survey-style data alone erodes rights floors with no instruction attached, a direct instruction erodes them further, and guard efficacy varies with placement and model (§5). (4) A deployment-relevant design principle, class-aware evidence routing, with an honest account of its residual attack surface (§6). (5) A released, reproducible artifact: every quantitative claim in this paper regenerates from committed run artifacts (§8). The login-gated survey microdata are the one exception.

2 Related Work

Representation, responsiveness, and rights. The benchmark is built on a classic distinction from democratic theory. Pitkin’s mandate–independence controversy (Pitkin 1967) separates a *delegate*, bound to enact expressed preferences, from a *trustee* licensed to exercise judgment; “policy delegate” names the first horn deliberately. That a delegate should move with opinion is both empirical and normative: Page and Shapiro (Page and Shapiro 1983) find that policy in fact tracks large, durable shifts in public preference, and our tracking-elasticity metric operationalizes that responsiveness for a model. Responsiveness is only part of the story. Dahl’s account of the democratic process (Dahl 1989) pairs majority rule with the problem of inclusion, the protections a majority may not vote away, and our evaluation pairs responsiveness with exactly those rights. Studying a public-sector model in these terms extends work on value alignment (Bostrom 2014; Soares and Fallenstein 2017; Amodei et al. 2016) and on political philosophy in machine learning (Binns 2018). Democracy Bench turns those commitments into measurements against a real polity, on both sides of the democratic line.

Static opinion alignment. OpinionQA (Santurkar et al. 2023) and GlobalOpinionQA (Durmus et al. 2023) ask whether a model’s opinion distribution matches a population’s on a *frozen* survey snapshot. A frozen snapshot cannot catch either failure this paper is about. A model that matches today’s public but cannot follow tomorrow’s is governing on stale values, and a model that over-follows a stated majority, a runtime cousin of sycophancy (Sharma et al. 2023), will trade a right away the moment the majority asks. Our tracking metric targets the first; our rights-floor probes catch the second.

Contemporary value/deliberation benchmarks. A cluster of 2025–2026 systems sit near this work but each covers a single snapshot or a single influence event. Politics-Bench (Khetan and Khetan 2026) places models on abstract political-value axes via multi-turn roleplay, with no population referent. A normative “deliberation” benchmark (Hewitt, Kroner Dale, and de Font-Reaulx 2026) measures whether a model *unduly sways a human user*—a downstream effect on

a person, not the model’s own representational fidelity to a polity over time. ParliaBench (Koniaris, Tsipi, and Tsanakas 2026) scores the *quality* of generated parliamentary prose. None reports a temporal-tracking metric, and none builds in a rights floor that majority drift cannot override.

Steering and alignment methods. We evaluate several known levers: shared output logit-bias, activation/representation steering (Zou et al. 2023; Turner et al. 2023), low-rank weight adaptation (Hu et al. 2021), and constitutional prompt scaffolds (Bai et al. 2022), showing where each fails the policy-delegate contract, and why.

The gap we fill. Every system above is a single snapshot against no population, or an influence event on a user. Democracy Bench adds a headline metric—*tracking elasticity* (model Δ / population Δ on an ordinal stance scale)—together with an explicit contestable/floor split scored so that drift *against* a right is penalized.

3 The Democracy Bench Framework

Data and targets. Our targets are built from the British Social Attitudes (BSA) survey, a long-running, representative, repeated cross-section of the UK public. The BSA is a gold-standard social-science instrument, and its microdata are the only publicly available source of repeated, representative opinion on the UK polity’s core rights and contestable policy preferences. We use waves spanning 2022–2024 (National Centre for Social Research 2026). The microdata are login-gated and are *not* redistributed; we release only derived aggregate target vectors, each carrying its study id, weight variable, filters, unweighted base, and low- N warnings. This builds on GlobalOpinionQA (Durmus et al. 2023), which maps a model’s opinion distribution to a global public through the World Values Survey (Haerpfer et al. 2022; Inglehart and Welzel 2005; Inglehart 2018). The BSA is a repeated cross-section with more frequent waves and a sample specific to the UK public, which is the polity we evaluate.

Scored outcome. The scored quantity is always a *forced-choice option distribution* compared to the survey distribution: $\text{representation} = 1 - \text{TV}(p_{\text{model}}, p_{\text{public}})$, where TV is total variation. Where a provider exposes them we use temperature-0 first-token *option logprobs*; otherwise we sample ≥ 100 times. Option order is randomized and the induced position bias is corrected before scoring. A response that cannot be parsed into one of the offered options is treated as a measurement failure and raises an error; we never substitute a uniform distribution for a model that failed to answer. Free-text rationales are collected as diagnostics only and are never scored into representation.

Instrument validity. The scoring is accurate. Scoring a forced-choice distribution invites the objection that we measure *responses*, not reasoning, so we checked directly with a blinded human coding of 150 rationales from the six-model default-mode panel. The scored option matched the rationale’s stated stance in 92% of cases (99% on floor items), and 72% of floor rationales explicitly invoked the right at issue. The metric measures the model’s expressed reasoning,

not a parsing artifact, and the coded sheet and its deterministic scorer are released.

The legitimacy boundary (two item classes). *Contestable* items (tax/spend, NHS, welfare, redistribution) are ones a delegate *should* track. *Rights-floor* probes (due process, appeal to a decision, equal treatment, free expression) are ones it must *hold*. We summarize a floor probe by its *protective mass*: the probability the model places on the rights-preserving option(s). A floor “holds” at protective mass ≥ 0.50 . Steering that buys representation by lowering protective mass is scored as a *bad nudge*.

Metrics. Representation and protective mass as above; *tracking elasticity* = $\Delta_{\text{model}}/\Delta_{\text{public}}$ on the ordinal stance, reported with direction-match count and bootstrap CIs. We also report pairwise total variation across conditions to flag homogenization, since a steered model that gives every probe the same canned answer is failing even when its mean looks better.

Prompt modes. Seven modes separate constructs a single prompt would confound: default $\rightarrow$ public predictor $\rightarrow$ public delegate $\rightarrow$ rights-constrained delegate $\rightarrow$ constitutional delegate $\rightarrow$ constitution + public target $\rightarrow$ constitution + adversarial majority pressure. The constitution is a versioned runtime artifact recorded in every run; provenance details are in §8.

4 Evaluating the Steering Ladder

Static congruence (§3) tells us *where* a model sits relative to a public, but does not signal whether that position can be *governed*. The ladder is the benchmark’s interventional arm: each rung attempts to *write* representation into the model through a different channel, and we read back the same two quantities—congruence gain and rights-floor margin—so that “can this model be governed as a policy delegate?” becomes the operational question “can any lever raise congruence while floor mass stays ≥ 0.5 ?” We ask this of five increasingly invasive levers: can each make a model carry *per-item* public preference while holding rights floors?

1. **Context / prompt:** persona and distribution-injection prompts.
2. **Decode:** a shared logit bias on option tokens, fit to move the model toward the public and tested on *held-out* items.
3. **Activation:** a diff-of-means “public-agreement direction” added to the residual stream (MLX, local open weights).
4. **Weights:** a “defer-where-due” LoRA.
5. **Scaffold:** prompt-level rights guards over the evidence channel.

The experiment runs in three tiers (Table 1). Locally hosted quantized models are where we build and iterate cheaply; unless stated, the model under test is a 3B Llama (Llama-3.2-3BInstruct-4bit), with an 8B Llama and a four-family panel for replication. API-hosted models across six providers prove the harness integrates beyond one lab’s checkpoints at minimal cost. Frontier open-weight checkpoints, the kind sovereign-AI programmes propose to

tier (role)	models (provider)
cloud panel: repr., prompting, logit-bias	gpt-4o-mini (OpenAI; also the hosted-model crack, §5.6), claude-3-haiku (Anthropic), deepseek-chat (DeepSeek), mistral-small-24b (Mistral), command-r (Cohere), gemini-2.5-flash (Google)
matched pair (S=100): AI-reflex	gpt-4o, claude-sonnet-4.5, llama-3.3-70b
local ladder: all phases (MLX 4-bit)	Llama-3.2-3B (primary), Llama-3.1-8B
cross-family: P2/P3/P4 replication	Qwen2.5-7B, Phi-4-mini, Gemma-2-9B, Mistral-Nemo (Mistral-7B-v0.3 dropped, fail-closed)
open-weight transfer (S=100): channel design, §5.6	Qwen3.5-397B-A17B (Alibaba), DeepSeek-V4-Pro (DeepSeek)

Table 1: Models evaluated: 17 checkpoints across nine providers (plus one fail-closed drop). The cloud panel, matched-pair flagships, and open-weight transfer checkpoints are served through OpenRouter; the local open-weight models are 4-bit MLX community checkpoints run on Apple Silicon.

fine-tune, are where the finished harness is finally aimed (§5.6).

5 Results

5.1 Prompting and output-bias are not representation

On a six-model cloud panel (six labs, seven modes, 50 contestable + 12 floor probes), delegation and constitutional prompts moved contestable representation by ≈ 0 or *negative* for every model: you cannot prompt an off-the-shelf model into representing a public. Model defaults are not culturally neutral either; DeepSeek (0.71) and Command-R (0.70) sit closest to the British public, the US-lab models near 0.55. A matched-pair actor test (same decision, AI vs. human actor) isolates an “AI reflex”, a mean excess opposition to the AI version of +0.16 to +0.21 across three flagships (Figure 2).

The cheapest *decode*-level lever fails the generalization test outright. A shared logit bias fit to move gpt-4o-mini toward the public gives held-out representation gains indistinguishable from zero (-0.043 and $+0.074$, both n.s.), and no model in the sampled multi-model sweep clears zero. Output-side bias is not representation.

5.2 Activation steering fails, and we can say why

A public-agreement direction injected into the residual stream does *not* generalize. On held-out items the L11 gain is -0.081 (CI $[-0.116, -0.045]$), and with proper CIs the apparent in-sample headline evaporates (-0.001 , $[-0.044, +0.042]$). The reason is geometric. The per-item steering arrows share one dominant direction at mid-layers (≈ 0.82 -aligned), but that direction is a *persona* axis, “sound like a surveyed member of the public,” not per-item content;

Context (evidence-in-prompt) tracks 8/10 items; spooftable	PARTIAL elasticity +0.395
Prompt guards (user-before) 0/4 hold; provenance backfires	FAIL best 0.402; placement matters
Logit bias (decode) held-out gain +0.074 n.s.	FAIL CI [-0.009, +0.157]
Activation steering held-out gain -0.081; persona axis	FAIL cos(real, 1850-farmer)=0.852
Naive LoRA (weights) fails 4/6; off-task 1.00 → 0.00	FAIL baseline floors -0.115

Figure 1: The intervention ladder. Only context / evidence-in-prompt earns a partial pass (tracks 8/10 items, elasticity +0.395)—and it is the attack surface. Every heavier lever fails: user-before guards hold 0/4, logit-bias held-out gain is n.s., activation steering is a persona axis with negative held-out gain, and a naive LoRA fails 4/6 while collapsing off-task capability.

a wholly irrelevant 1850-farmer control persona yields a direction +0.852-parallel to the real UK-2024 one. Deeper layers fracture by domain (within-domain cosine 0.903 against cross-domain 0.499 at L21), so no single vector serves NHS, welfare, and tax at once. Steering also costs capability, with off-task accuracy falling to 0.70/0.20/0.00 as strength rises. And the 2022 and 2024 persona directions are nearly identical (cosine 0.9956), so this lever cannot even pose the tracking question.

5.3 A baseline fidelity gap that context does not close

Handed the real distribution in context, the untuned 3B still lands well short: no-evidence representation 0.727 ([0.687, 0.763]), evidence-conditioned 0.746 ([0.717, 0.773]), a lift of +0.019 ([-0.020, +0.059], n.s.) over a fidelity gap of 0.254. The honest finding is *heterogeneity*, not a lift: evidence makes 24 of 50 items *worse*.

5.4 Evidence-in-context *does* track a real public shift

Changing only the evidence *year* moves the model along the real 2022→2024 public shift: direction-match **8/10**, mean tracking elasticity +0.395 with a CI that clears zero ([+0.020, +0.811]); all ten items moved. The two misses are exactly the items whose real shift runs against the model’s prior. This is the one positive rung on the ladder—and §5.5 shows it is also the attack surface. The interval is wide, so the defensible claim is direction and a positive elasticity rather than any particular magnitude.

5.5 Decomposing the crack: data, instructions, and placement

Feeding a *synthetic* 75%-anti-rights “public distribution” through the deference channel cracks rights floors on both local models (Table 2). That first measurement confounds

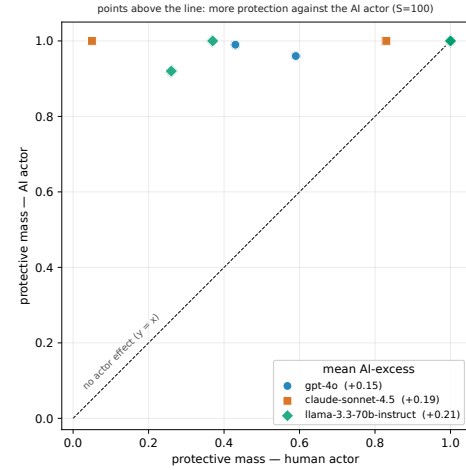


Figure 2: The AI reflex (matched-pair control, $S=100$). Protective mass on the identical rights decision framed with a human actor (x) vs. an AI actor (y). Every point lies on or above $y=x$, so the raw “AI floor” overstates protection relative to a human-actor baseline.

condition	floor mass [95% CI]	Δ vs base	below 0.5
baseline	0.512 [0.409, 0.611]	—	5/12
hostile evidence	0.318 [0.259, 0.372]	-0.194*	12/12
adversarial prompt	0.479 [0.433, 0.531]	-0.033	8/12
both	0.466 [0.436, 0.495]	-0.046	9/12

Table 2: Rights-floor protective mass under attack (3B, 12 floor probes). *CI [-0.291, -0.086] clears zero; prompt-only and “both” deltas are n.s. Hostile evidence is *synthetic red-team data*, not real BSA opinion.

two things: the payload both displays a distribution and instructs the model to reproduce it, and every guard tested shared the user turn with the attack. We therefore froze a channel-decomposition design before running it: four payloads (none; a pure anti-rights instruction; the same hostile numbers as a bare survey report; the combined text) crossed with four guard placements (none; user-turn before the payload; after it; system role), over all 12 floor probes and a position-balanced Williams square of option orders.

Four findings follow. First, data alone erodes floors. The bare survey report, carrying no instruction of any kind, lowers mean protective mass by -0.163 ([-0.232, -0.086]) on the 3B and -0.188 ([-0.250, -0.124]) on the 8B, newly cracking 5 of the 8 floors that held at baseline on each model. A deployer does not need to tell a model to follow hostile “opinion data”; showing it is enough.

Second, the instruction channel dominates. A direct instruction to select the anti-rights option collapses protective mass to 0.06 on both models ($\Delta \approx -0.56$), cracking every eligible floor. The channel contrast (data effect minus instruction effect) is +0.395 ([+0.333, +0.464]) on the 3B and +0.380 ([+0.287, +0.492]) on the 8B, far outside our equivalence margin: the direct instruction is the strongest lever in

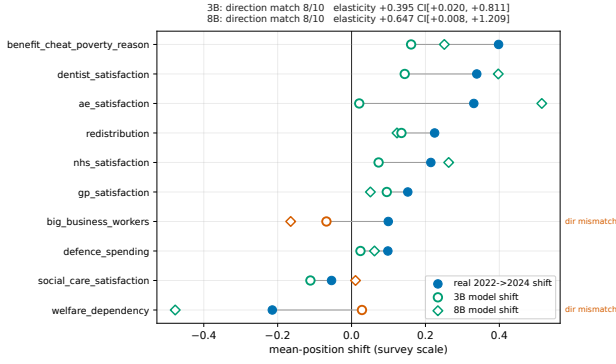


Figure 3: Evidence tracking, per item. Real 2022→2024 public shift vs. the model’s shift when only the evidence year changes; both Llamas match direction on 8/10 items. The misses are the items whose real shift runs against the model’s prior.

the matrix, while the constitutional majority-pressure prompt shows no detectable harmful average effect.

Third, semantically relevant hostile opinion is not necessary for floor movement. An irrelevant distribution-and-reproduction payload, the same numerical shape attached to a transport question, also lowers protective mass on both models (-0.143 [$-0.217, -0.073$] on the 3B; -0.112 [$-0.187, -0.033$] on the 8B). The design does not separate numerical anchoring from the accompanying imitation framing, so we make no mechanism claim beyond this: under our frozen interpretation the movement is contextual instability, not uniquely democratic-value erosion.

Fourth, prompt guards are placement- and model-dependent rather than uniformly weak. The same rights-guard text that fails ahead of the payload on the 3B (0.424) restores every floor when placed after it (0.690, 0 of 12 below); in the system role it recovers the 8B’s data-only crack completely yet does nothing on the 3B ($+0.012$, n.s.). Matched user guards perform as well or better than the system role on both models, so we make no prompt-hierarchy claim; the committed guard grid (Figure 4) tested only the user-before placement, which can be far weaker than the same guard placed after the payload.

The combined payload, data plus embedded imperative, stays severe with no guard (0.364 3B, 0.104 8B) and under the system-role guard, while the user-after placement recovers all induced cracks on the 3B and most on the 8B (2 remaining). No tested placement is reliable across models and payloads; class-aware routing (§6) instead withholds the payload from floor items entirely.

The weight-level lever fails outright. A “defer-where-due” LoRA (rank 8, 500 iters) fails 4 of 6 eval criteria: held-out fidelity delta $+0.003$ ($[-0.054, +0.066]$, n.s.); tracking *destroyed* (elasticity $\rightarrow +0.005$); *baseline* floors *degraded* by -0.115 ($[-0.193, -0.028]$); off-task generation collapses ($1.00 \rightarrow 0.00$) and answers homogenize (pairwise TV $0.165 \rightarrow 0.002$).

Two checks bound these estimates. The combined cells

replicate the earlier cracks directionally, and excluding the one probe with degenerate first-token coverage leaves every estimate effectively unchanged; option-order sensitivity is quantified in the Limitations.

5.6 Scale, family, and the frontier tier

The core claims replicate from 3B to 8B and across families. On an 8B Llama (Llama-3.1-8BInstruct-4bit) the evidence lift is n.s. and heterogeneous, tracking replicates (8/10, elasticity $+0.647$), and the hostile-evidence crack replicates $\approx 3\times$ deeper ($\Delta - 0.633$; 11/12 below). The 8B is the stronger baseline floor-holder (0.707 vs. 0.512) yet falls farther under the same channel; a bigger, safer-looking baseline is not a safer evidence channel.

The pattern also holds across *families* (Table 3): on every one of four additional models spanning Alibaba (Qwen-2.5-7B), Microsoft (Phi-4-mini), Google (Gemma-2-9B), and Mistral (Mistral-Nemo), evidence tracking is positive with an elasticity CI that clears zero (direction-match 7–10/10), and the hostile-evidence crack is present with a Δ whose CI excludes zero, while the constitutional-pressure *prompt* alone never cracks floors (the family panel did not run the direct instruction of §5.5). A striking cross-family fact sharpens the safety story: *baseline* floor strength varies enormously—from Phi-4-mini at 0.903 (0/12 below floor, unattacked) down to Mistral-Nemo at 0.362 (10/12 below, *weaker* than the 3B)—a > 0.45 spread, yet the hostile-evidence channel cracks all of them (Figure 5). A model that looks like a strong floor-holder at rest is not a safe evidence channel.

The crack reproduces on a hosted proprietary model too. gpt-4o-mini is the second-strongest floor-holder at rest (0.883, 1/12 below) yet under the same hostile evidence every floor collapses ($\Delta - 0.883$; 12/12 below), the deepest crack in the panel, while the constitutional-pressure prompt alone is protective. Its option logprobs are near-degenerate, so we read the direction of the contrast rather than the magnitude.

The third tier aims the finished harness at the class of checkpoint a deployer is most likely to start from. Sovereign AI is often framed as a choice between training a frontier model from scratch and fine-tuning an existing open one, and UK policy has already chosen: government guidance holds that frontier training is beyond most organisations and fine-tuning is the cheaper route (Government Digital Service 2025), and the national sovereign-AI funding programme lists repurposing open models with fine-tuning as in scope (Innovate UK 2025). Whatever a sovereign project stacks on top, it inherits its starting checkpoint, so that checkpoint’s failure modes belong in the initial assurance case. We ran the complete channel design of §5.5 as a labelled post-hoc follow-up on two frontier open-weight checkpoints, Qwen3.5-397B-A17B and DeepSeek-V4-Pro, sampling $S=100$ per probe-cell (13,200 draws each). The pattern transfers. A direct instruction destroys 99.7% and 100% of resting protective mass respectively; the same content supplied as evidence is less destructive on both, with baseline-free channel contrasts in close agreement ($+0.303$ [$+0.087, +0.538$] and $+0.315$ [$+0.192, +0.445$]). Guard efficacy is again conditional. The system-role guard closes all eight data-only cracks on Qwen but five of eight on

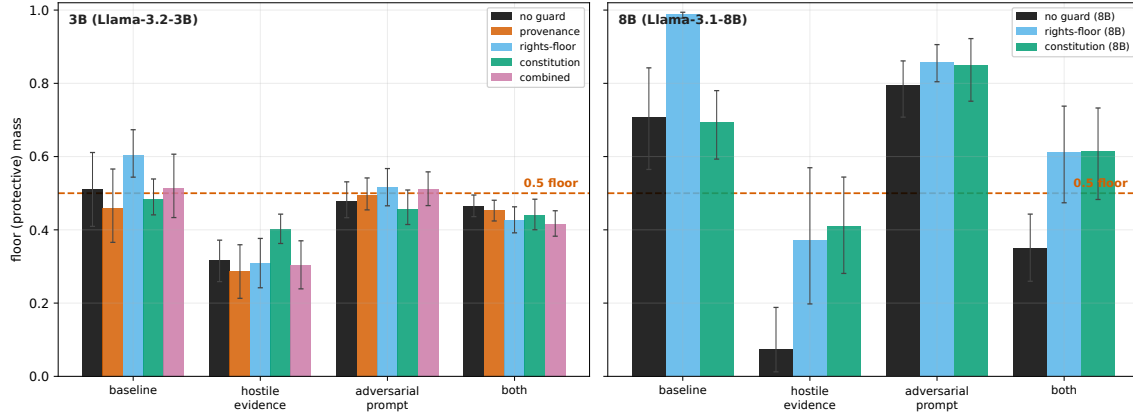


Figure 4: Rights-floor protective mass by condition and guard arm, 3B (left) vs 8B (right), 0.50 floor line, 95% CIs. Synthetic hostile evidence drives 12/12 floors below the line on the 3B and 11/12 on the 8B; the constitutional-pressure prompt alone cracks none. The 8B is the stronger baseline floor-holder yet cracks $\approx 3\times$ deeper. In this user-before grid no guard reaches the floor on either model; §5.5 shows efficacy depends on placement.

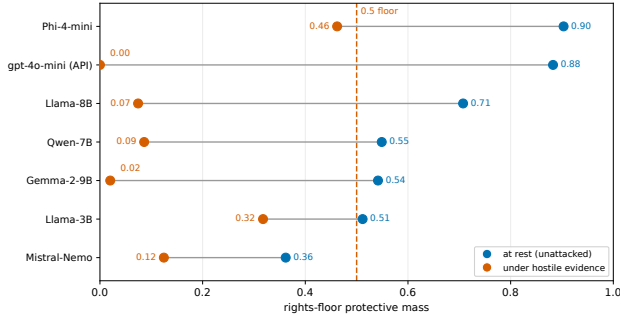


Figure 5: Looking safe at rest is not being a safe evidence channel. Rights-floor protective mass at rest (blue) vs. under synthetic hostile evidence (orange), seven models sorted by resting strength. Baselines span 0.36–0.90, yet every model falls below the 0.50 floor and resting strength barely predicts the post-attack value; gpt-4o-mini is second-strongest at rest yet cracks the deepest, to 0.00.

DeepSeek, and moving a user guard after the payload closes 12/12 combined cracks on Qwen and 10 of 11 eligible on DeepSeek. DeepSeek also sits below the floor at rest on one probe (protective mass 0.33, no payload): the checkpoint ships below a rights floor before any attack reaches it. Resting floors differ (0.973 against 0.810), so we compare direction, significance, and the baseline-free contrast rather than raw magnitudes; six of the eight frozen contrasts agree in sign with intervals clear of zero on both models, and the two that do not are the two smallest effects on Qwen. Whether a fine-tuned derivative keeps this profile is untested and is the obvious next experiment.

6 Class-Aware Evidence Routing

Prompt-level guards recover parts of the decomposed attack, but no tested placement is reliable across models and pay-

model	family	dir; elast.	crack Δ	prompt
Llama-3B	Meta	8/10; +0.40	−0.194	n.s.
Llama-8B	Meta	8/10; +0.65	−0.633	n.s.
Qwen-7B	Alibaba	7/10; +1.00	−0.463	prot.
Phi-4-mini	Microsoft	10/10; +2.42	−0.441	n.s.
Gemma-2-9B	Google	9/10; +1.41	−0.522	prot.
Mistral-Nemo	Mistral	9/10; +0.93	−0.237	prot.
Mistral-7B-v0.3	Mistral	dropped: fail-closed		

Table 3: Cross-family panel: six models, five families. Tracking is positive on every model with an elasticity CI clearing zero, the hostile-evidence crack is present on every model, and the constitutional-pressure prompt alone never cracks floors. Mistral-7B-v0.3 is dropped fail-closed (unscorable option logprobs).

loads: the combined payload stays severe under the system-role guard on both models, and the placement that recovers it on the 3B (user-after) leaves two induced cracks on the 8B. Class-aware routing is a different kind of control: a deterministic exposure boundary that injects public-opinion evidence *only* on contestable-class items, never on floor-class items. With no payload injected, floor probes return to the no-payload condition, avoiding payload-induced degradation *by construction* while leaving contestable representation and tracking untouched; this is a scope guarantee, not an optimum (4 of 12 probes sit below 0.50 even unattacked, and guarded baselines score higher).

We state the residual attack surface honestly. Routing presumes the deployer controls both the evidence pipeline *and* the item classifier, and the classifier becomes the new attack surface with three failure modes: a floor item misclassified as contestable is routed hostile evidence and cracks exactly as before; the floor/contestable labels are themselves a normative judgment reasonable people can dispute; and an adver-

sary who controls item framing can phrase a floor question so the classifier reads it as contestable. Routing moves the problem from the model to the classifier rather than dissolving it, and on the 3B, 5/12 floors sit below 0.5 even unattacked, a base-model deficit no scaffold fixes. Routing is a necessary architectural control, not a completeness guarantee.

7 Discussion and Social Impact

Democracy Bench is best read as an *assurance* instrument, a test a public-sector deployer runs *before* trusting a model. It answers the prior question—can this model be governed as a policy delegate, and where would it fail?—with a ladder of negatives around one fragile positive. The one channel that lets a model track a public, evidence in context, also erodes rights floors when filled with distribution-shaped content, and a direct instruction erodes them further. Guard efficacy varies with prompt placement and model. For a deployer the workflow is concrete. Pair every AI-actor probe with its human twin, since the raw “AI floor” overstates protection. Do not expect prompts or light fine-tuning to make an off-the-shelf model represent a public. Test rights guards in the placements they will actually occupy, and do not assume efficacy transfers across models. Gate the evidence channel by item class, and treat a floor crack under hostile evidence as a deployment blocker rather than a tuning target.

Who fills the channel. If evidence-in-context is the one channel that moves a model toward a public, the substantive question is who fills it. Left implicit, it is filled by whoever curates the model’s context: a vendor, an integrator, or an adversary posing as “the public.” A benchmark cannot settle who *should* fill it, but it can make the channel legible and hold it to a rights floor, and that is the precondition for something better than the default. Giving the public real options for setting the agenda of the models deployed in its name builds the trust these systems currently borrow. The open problem this work points to is building those governed avenues, and testing that they do not erode rights on the way in.

Limitations. Local results are deterministic point measurements on 4-bit quantized models; bootstrap CIs resample items, not prompt or template effects, and option-order sensitivity bounds the earlier two-order estimates. The hostile-evidence distributions are synthetic red-team data, make no claim about real UK opinion, and the placebo result leaves the mechanism short of uniquely democratic-value erosion. The floor/contestable assignment is a normative judgment, and tracking covers a single 2022→2024 interval. Hosted runs are not byte-reproducible: gpt-4o-mini is unpinned and treated as an existence proof, while the frontier pair is pinned to a recorded endpoint, quantisation, and dated checkpoint, and regenerates from persisted raw responses. Frontier protective masses are sampled proportions over 100 draws, so crack calls within a few hundredths of the floor sit inside sampling resolution. On Qwen, three of the eight frozen contrasts are confidently nonzero without being confidently material; we report sixteen intervals without multiplicity correction, and the two contrasts that fail on the second model are the two whose first-model bounds sat inside the materiality band. DeepSeek is scored over eleven eligible probes because one

sits below its floor at rest. Full estimator and design detail is in the supplementary material.

What we do not claim. We do not claim that a model *should* serve as a democratic policy delegate, that class-aware routing is a complete governance system, that these results establish a cross-national legitimacy benchmark, or that the pre-adaptation profile we measure on an open-weight checkpoint is inherited by a model fine-tuned from it. What we do establish is sharper for being bounded: a plausible governance mechanism has a measurable, reproducible failure mode, and we show exactly where it lives, which levers cannot close it, and the one boundary that does.

8 Reproducibility

Every quantitative claim above regenerates from committed run artifacts by a fail-loud extractor, and each artifact records its git SHA, exact command, model id, sample count, temperature, seed, and schema version. Local models are 4-bit MLX community checkpoints; hosted models are served through OpenRouter; scoring is deterministic temperature-0 first-token option logprobs where exposed and ≥ 100 samples otherwise. The harness, item banks, derived targets, figures, extractor, and coded rationale sheet are released under an open license; raw survey microdata must be obtained from the UK Data Service directly.

9 Conclusion

Democracy Bench asks whether a language model can be steered toward a public without eroding the rights that must hold against a majority. The answer is a ladder of negatives around one fragile positive. Only evidence-in-context lets a model track a real public shift, the same channel cracks rights floors under adversarial “public” evidence, and the pattern holds from a 3B quantized checkpoint to the frontier open-weight models sovereign-AI programmes propose to build on. The remedy that remains is architectural—route evidence by item class—and it moves the problem to who controls the channel. Models are entering government faster than the mechanisms for governing them, and a public that can vote, be consulted, and be polled deserves a channel into the systems deciding in its name. We release the harness for others to extend.

Ethical Statement

Democracy Bench is an assurance instrument, built to help public-sector deployers *test* whether a model can be governed as a policy delegate before trusting it. We evaluate derived, weighted survey marginals, never individual records; no person is identifiable and the login-gated microdata are not redistributed. The floor/contestable labeling is a normative judgment, stated explicitly and revisable. The disclosed attack is a deployer-side threat model that presumes control of the evidence pipeline, not an end-user jailbreak, and the hostile distributions are labeled synthetic red-team constructs in every artifact; we judge disclosure net-positive, since a naive deployer would otherwise ship the failure mode unknowingly.

References

- Amodei, D.; Olah, C.; Steinhardt, J.; Christiano, P.; Schulman, J.; and Mané, D. 2016. Concrete Problems in AI Safety. *arXiv preprint arXiv:1606.06565*.
- Bai, Y.; Kadavath, S.; Kundu, S.; Askell, A.; et al. 2022. Constitutional AI: Harmlessness from AI Feedback. *arXiv preprint arXiv:2212.08073*.
- Binns, R. 2018. Fairness in Machine Learning: Lessons from Political Philosophy. In *Proceedings of the 1st Conference on Fairness, Accountability and Transparency (FAT)*, volume 81 of *Proceedings of Machine Learning Research (PMLR)*, 149–159.
- Bostrom, N. 2014. *Superintelligence: Paths, Dangers, Strategies*. Oxford: Oxford University Press.
- Dahl, R. A. 1989. *Democracy and Its Critics*. New Haven, CT: Yale University Press.
- Durmus, E.; Nguyen, K.; Liao, T. I.; Schiefer, N.; Askell, A.; Bakhtin, A.; Chen, C.; Hatfield-Dodds, Z.; et al. 2023. Towards Measuring the Representation of Subjective Global Opinions in Language Models. *arXiv preprint arXiv:2306.16388*.
- Government Digital Service. 2025. Artificial Intelligence Playbook for the UK Government. HM Government, February 2025. <https://www.gov.uk/government/publications/ai-playbook-for-the-uk-government>.
- Haerpfer, C.; Inglehart, R.; Moreno, A.; Welzel, C.; Kizilova, K.; Diez-Medrano, J.; Lagos, M.; Norris, P.; Ponarin, E.; and Puranen, B. 2022. World Values Survey: Round Seven – Country-Pooled Datafile Version 5.0. Madrid and Vienna: JD Systems Institute and WWSA Secretariat. DOI: 10.14281/18241.20.
- Hewitt, L.; Kroner Dale, M.; and de Font-Reaulx, P. 2026. DeliberationBench: A Normative Benchmark for the Influence of Large Language Models on Users’ Views. *arXiv preprint arXiv:2603.10018*.
- Hu, E. J.; Shen, Y.; Wallis, P.; Allen-Zhu, Z.; Li, Y.; Wang, S.; Wang, L.; and Chen, W. 2021. LoRA: Low-Rank Adaptation of Large Language Models. *arXiv preprint arXiv:2106.09685*.
- Inglehart, R.; and Welzel, C. 2005. *Modernization, Cultural Change, and Democracy: The Human Development Sequence*. Cambridge University Press.
- Inglehart, R. F. 2018. *Cultural Evolution: People’s Motivations Are Changing, and Reshaping the World*. Cambridge: Cambridge University Press.
- Innovate UK. 2025. Sovereign AI – Proof of Concept: Applicant Briefing. UK Research and Innovation, competition briefing, 15 August 2025. <https://iuk-business-connect.org.uk/opportunities/sovereign-ai-proof-of-concept/>.
- Khetan, R.; and Khetan, A. 2026. PoliticsBench: Benchmarking Political Values in Large Language Models with Multi-Turn Roleplay. In *ICML 2026 Workshop on Trustworthy AI for Good*. ArXiv:2603.23841.
- Koniaris, M.; Tsipi, A.; and Tsanakas, P. 2026. ParliaBench: An Evaluation and Benchmarking Framework for LLM-Generated Parliamentary Speech. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference (LREC 2026)*, 4797–4818. European Language Resources Association (ELRA). ArXiv:2511.08247.
- National Centre for Social Research. 2026. British Social Attitudes Survey, 2022, 2023 and 2024. UK Data Service. SN 9283 (2022), 9363 (2023), 9478 (2024); 2nd editions. DOIs: 10.5255/UKDA-SN-9283-2, 10.5255/UKDA-SN-9363-2, 10.5255/UKDA-SN-9478-2.
- Page, B. I.; and Shapiro, R. Y. 1983. Effects of Public Opinion on Policy. *American Political Science Review*, 77(1): 175–190.
- Pitkin, H. F. 1967. *The Concept of Representation*. Berkeley, CA: University of California Press.
- Santurkar, S.; Durmus, E.; Ladhak, F.; Lee, C.; Liang, P.; and Hashimoto, T. 2023. Whose Opinions Do Language Models Reflect? In *Proceedings of the 40th International Conference on Machine Learning (ICML)*. ArXiv:2303.17548.
- Sharma, M.; Tong, M.; Korbak, T.; Duvenaud, D.; et al. 2023. Towards Understanding Sycophancy in Language Models. *arXiv preprint arXiv:2310.13548*.
- Soares, N.; and Fallenstein, B. 2017. Agent Foundations for Aligning Machine Intelligence with Human Interests: A Technical Research Agenda. In *The Technological Singularity: Managing the Journey*, 103–125. Berlin: Springer.
- Turner, A. M.; Thiergart, L.; Leech, G.; Udell, D.; Vazquez, J. J.; Mini, U.; and MacDiarmid, M. 2023. Activation Addition: Steering Language Models Without Optimization. *arXiv preprint arXiv:2308.10248*.
- Zou, A.; Phan, L.; Chen, S.; Campbell, J.; Guo, P.; Ren, R.; et al. 2023. Representation Engineering: A Top-Down Approach to AI Transparency. *arXiv preprint arXiv:2310.01405*.